



Project Introduction

This project is about handwritten digit recognition using a Convolutional Neural Network (CNN) trained .on the MNIST dataset

The goal is to build a model that can correctly classify .handwritten digits from 0 to 9 with high accuracy

Name: Jawad Alghelfes

Supervisor: Eng. Badr Al-Barrak

School: Technical Gifted Secondary School

أكاديمية طويق
Tuwaiq Academy



Project Report: Handwritten Digit Recognition (MNIST)

Phase 1 – Project Planning Problem Statement:

We want to automatically recognize handwritten digits to reduce manual data entry and improve processing speed for document-heavy organizations.

Success Metrics:

Achieve a Final Test Accuracy greater than 99%, enabling 95% of handwritten forms to be processed automatically.

Data Information:

MNIST dataset consisting of 70,000 grayscale images (28×28 pixels) split across 10 classes (digits 0–9).

Risks & Mitigations:

Overfitting (monitored via validation sets).

Confusable digits like 4 and 9 (mitigated by using spatial CNN filters).

Varied handwriting styles (mitigated by a large training pool).

Phase 2 – Model Training & Architecture Data Preprocessing:

Pixel values were normalized from $[0, 255]$ to $[0, 1]$ for faster convergence.

A single-channel dimension was added $(28, 28, 1)$, and data was strictly partitioned into Train, Validation, and Test sets.

CNN Model Layers:

Conv2D (32) → MaxPooling2D → Conv2D (64) → MaxPooling2D → Flatten → Dense (64, ReLU) → Dense (10, Softmax).

Hyperparameters:

Trained using the Adam optimizer and Sparse Categorical Crossentropy loss function for 5 epochs with a batch size of 32.

Phase 3 & 4 – Results & Analysis

Final Metrics:

Training Accuracy: 99.43% | Validation Accuracy: 98.93% | Test Accuracy: 99.06% 🌟 (Target met).

Experiment Table:

Model Description	Validation Accuracy	Test Accuracy	Status
Baseline CNN	98.93%	99.06%	SUCCESS

Core Analysis:

The model exceeded the 99% accuracy threshold. Visual analysis via the Confusion Matrix and Error Analysis images proves that remaining errors only occur when numbers are severely distorted or ambiguous even to human readers.

Limitations & Next Steps:

The model is optimized for clean MNIST data and might degrade on noisy, real-world documents. Future steps include introducing Data Augmentation and testing deeper CNN models.

Phase 5 – Reflection

What Worked:

The chosen CNN architecture successfully extracted robust geometric features, leading to high classification accuracy.

Challenges:

Handling highly ambiguous or messy human handwriting styles where lines overlap.

Key Learnings:

Gained hands-on experience in full end-to-end ML pipelines including: feature scaling, spatial image resizing, tracking evaluation charts, and saving model states (.h5) for final deployment.